Chapter 9. Discriminative Methods: Logistic Regression

Lecture 7-2 (discriminative vs generative, logit model, GLM, maximum likelihood, confounding, extensions, multinomial regression)

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1 Discriminative Methods versus Generative Models

Discriminative: model $\Pr(Y = c_k \mid X = x)$ directly	Generative: model (X, Y), then derive $\Pr(Y = c_k \mid X = x)$
K-nearest neighbors, logistic regression, classification trees, ensembles (bagging, random forest, boosting), support vector machines	linear and quadratic discriminant analysis, naive Bayes, Gaussian mixture models

Examples of classification: tumour cells benign/malignant; credit-card transactions legitimate/fraudulent; news stories finance/weather/entertainment/sports. We observe (X, Y) with Y a class label and learn $C(X)$ that predicts Y; often we estimate $P(Y = k \mid X)$ and classify to the most probable class.

2 Why Not Linear Regression?

Code $Y = 0$ (No) and $Y = 1$ (Yes) and fit $\hat{Y} = \beta_0 + \beta X$, classifying by $\hat{Y} > 0.5$? There is a model mismatch: $\mathbb{E}[Y \mid X] = P(Y = 1 \mid X)$ must lie in $[0, 1]$, but a linear model is unbounded and can give invalid probabilities; the relationship is typically nonlinear in X. (For two classes, the least-squares fit is in fact equivalent to LDA for the classification rule, but the probability estimates are poor.)

With more than two classes, coding e.g. $Y = 1$ (stroke), 2 (drug overdose), 3 (epileptic seizure) imposes an ordering and equal spacing that has no meaning, and different codings give different fits. Multiclass logistic regression or discriminant analysis is appropriate.

3 The Logistic Model

For two classes $Y \in \{c_1, c_2\}$, model $\Pr(Y = c_1 \mid X)$ through the **logit** transformation:

$$\log \frac{\Pr(Y = c_1 \mid X)}{\Pr(Y = c_2 \mid X)} = \beta_0 + \beta_1 X,$$

$$\Pr(Y = c_1 \mid X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}, \quad \Pr(Y = c_2 \mid X) = \frac{1}{1 + e^{\beta_0 + \beta_1 X}},$$

which keeps probabilities in $[0, 1]$ and gives an S-shaped curve. β_1 is the change in the **log odds** for a one-unit increase in X; e^{β_1} multiplies the odds; the change in the probability itself depends on X, $\partial p / \partial X = \beta_1 p(1 - p)$.

3.1 Logistic regression as a GLM

A generalized linear model consists of a random component (Y from an exponential family), a linear predictor $\eta = X^\top \beta$, and a link $g(\mathbb{E}[Y]) = \eta$. Logistic regression has $Y \sim \text{Bernoulli}(p)$ and the logit link $g(p) = \log \frac{p}{1-p}$, which is the canonical link of the Bernoulli family (the natural parameter of $p^y (1 - p)^{1-y} = \exp\{y \log \frac{p}{1-p} + \log(1 - p)\}$). Hence $\log \frac{p}{1-p} = X^\top \beta$.

4 Estimation by Maximum Likelihood

Motivation. Logistic regression has no closed form, so coefficients are found by maximizing the likelihood; the standard errors come from its curvature.

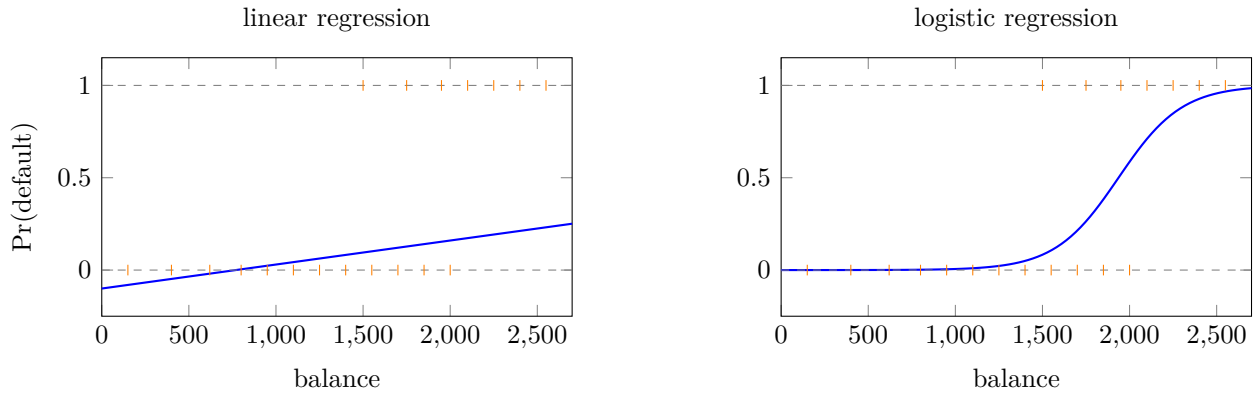


Figure 1: Default data (schematic points; orange ticks are $Y = 0$ at the bottom and $Y = 1$ at the top). A linear fit can give probabilities below 0 (and above 1); the fitted logistic curve $\hat{p} = e^{-10.65+0.0055X}/(1 + e^{-10.65+0.0055X})$ stays in $[0, 1]$.

The likelihood of the observed zeros and ones is

$$\mathcal{L}(\beta_0, \beta) = \prod_{i:y_i=1} p(x_i) \prod_{j:y_j=0} \{1 - p(x_j)\}, \qquad \ell(\beta_0, \beta_1) = \sum_i \Big[y_i(\beta_0 + \beta_1 x_i) - \log\left(1 + e^{\beta_0+\beta_1 x_i}\right) \Big].$$

Theorem: score equations and concavity

With $p_i = p(x_i)$ and design rows $\tilde{x}_i = (1, x_i)^\top$,

$$\nabla \ell = \sum_i (y_i - p_i)\tilde{x}_i = \tilde{X}^\top (y - p), \qquad \nabla^2 \ell = -\tilde{X}^\top W \tilde{X}, \; W = \mathrm{diag}\{p_i(1 - p_i)\}.$$

ℓ is concave, so the MLE solves $\tilde{X}^\top(y-p) = 0$ and is found by Newton–Raphson, equivalently **iteratively reweighted least squares**: $\beta^{\text{new}} = (\tilde{X}^\top W \tilde{X})^{-1} \tilde{X}^\top W z$ with working response $z = \tilde{X}\beta + W^{-1}(y - p)$.

Proof [Supplement]. $\frac{d}{d\eta} \log(1 + e^\eta) = \frac{e^\eta}{1+e^\eta} = p$ and $\frac{dp}{d\eta} = p(1 - p)$; differentiate termwise. The Hessian is negative semidefinite. A Newton step $\beta + (\tilde{X}^\top W \tilde{X})^{-1} \tilde{X}^\top (y - p)$ rearranges into the weighted least-squares form. ■

The asymptotic covariance of $\hat{\beta}$ is $(\tilde{X}^\top \hat{W} \tilde{X})^{-1}$ (inverse Fisher information), which gives the standard errors and z-statistics. If the classes are perfectly separable, the likelihood increases without bound as $\|\beta\| \to \infty$ and the MLE does not exist (estimates become unstable — see also Ch. 10).

5 Logistic Regression on the Default Data

	Coefficient	Std. error	z-statistic	p-value
Intercept	−10.6513	0.3612	−29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

A one-unit increase in balance increases the log odds of default by 0.0055. The z-statistic tests H_0: no relationship between X and the odds, i.e. $H_0 : \beta_1 = 0$; here $p < 0.0001$.

Predictions.

$$\hat{p}(1000) = \frac{e^{-10.6513+0.0055\times 1000}}{1 + e^{-10.6513+0.0055\times 1000}} = 0.006, \qquad \hat{p}(2000) = \frac{e^{-10.6513+0.0055\times 2000}}{1 + e^{-10.6513+0.0055\times 2000}} = 0.586.$$

A qualitative predictor. With student status: intercept -3.5041, student[Yes] 0.4049 ($p = 0.0004$), so $\Pr(\text{default} \mid \text{student}) = e^{-3.0992}/(1+e^{-3.0992}) = 0.0431$ and $\Pr(\text{default} \mid \text{non-student}) = e^{-3.5041}/(1+e^{-3.5041}) = 0.0292$.

6 Multiple Logistic Regression and Confounding

$$\log \frac{p(X)}{1-p(X)} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p, \qquad p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}.$$

	Coefficient	Std. error	z-statistic	p-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income (thousands)	0.0030	0.0082	0.37	0.7115
student[Yes]	-0.6468	0.2362	-2.74	0.0062

Why is the student coefficient now *negative* when it was positive alone? **Confounding**: students tend to carry higher balances, so their marginal default rate is higher; but at each fixed balance, students default *less* than non-students. Multiple logistic regression teases the two apart: the marginal association mixes the direct (negative) effect with the indirect path through balance.

7 Extensions of Logistic Regression

- **Polynomial logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d$. Modeling the event $\texttt{wage} > 250$ with a degree-4 polynomial in age gives the fitted probability with 95% bands (wide where data are scarce).
- **Piecewise-constant logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i)$ with step functions of age.
- **GAM for classification**: $\log \frac{P(X)}{1-P(X)} = \beta_0 + f_1(X_1) + \cdots + f_p(X_p)$; for the Wage data, $\beta_1\,\texttt{year} + f_2(\texttt{age}) + f_3(\texttt{education})$ with a smoothing spline (5 df) in age and a step function for education.
- **Penalized logistic regression**: minimize $-\log \mathcal{L}(\beta_0, \beta) + \lambda\|\beta\|_2^2$ (ridge) or $+\lambda\|\beta\|_1$ (lasso), generally $+\lambda\|\beta\|_q^q$; the penalty also gives finite estimates under separation.
- **Multinomial logistic regression** (below).

8 More Than Two Classes: Multinomial Regression

With K classes (e.g. stroke, drug overdose, epileptic seizure), choose class K as baseline and model, for $k = 1, \ldots, K-1$,

$$\log \frac{\Pr(Y=k \mid X=x)}{\Pr(Y=K \mid X=x)} = \beta_{k0} + \beta_{k1} x_1 + \cdots + \beta_{kp} x_p.$$

Theorem: implied probabilities

$$\Pr(Y=k \mid X=x) = \frac{e^{\beta_{k0}+\beta_k^\top x}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_l^\top x}} \quad (k<K),$$

$$\Pr(Y = K \mid X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0} + \beta_l^\top x}},$$

and the log odds between any two classes k, k' is linear: $\log \frac{\Pr(Y=k|x)}{\Pr(Y=k'|x)} = (\beta_{k0} - \beta_{k'0}) + (\beta_k - \beta_{k'})^\top x$.

Proof [Supplement]. Write $\pi_k = \Pr(Y = k \mid x) = \pi_K e^{\eta_k}$ for $k < K$; the probabilities sum to 1, so $\pi_K(1 + \sum_l e^{\eta_l}) = 1$. The difference of two log odds against K gives the pairwise log odds. ■

The choice of baseline does not affect the fitted probabilities, only the parametrization. A symmetric version (softmax) uses $\Pr(Y = k \mid x) = e^{\eta_k} / \sum_{l=1}^{K} e^{\eta_l}$ with one set of coefficients per class (identifiable only up to a common shift).

Summary

- Discriminative methods model $\Pr(Y \mid X)$ directly; generative methods model $\Pr(X \mid Y)\Pr(Y)$.
- Logistic regression models the log odds linearly (a Bernoulli GLM with logit link) and is fitted by maximum likelihood (IRLS).
- Coefficients are changes in log odds; multiple regression reveals confounding (student effect reverses sign given balance).
- Polynomial, step-function, GAM and penalized versions extend it; multinomial regression handles $K > 2$ classes with linear pairwise log odds.

9 Selected exercises

Motivation. Two facts about the logistic likelihood — non-existence of the MLE under separation and the closed form for a binary predictor — followed by ISLR Exercises 4.6 and 4.12 on reading coefficients and on the softmax parametrization.

Exercise 9.1 (No MLE under complete separation)

Let $\ell(\beta) = \sum_i [y_i \tilde{x}_i^\top \beta - \log(1 + e^{\tilde{x}_i^\top \beta})]$ with $y_i \in \{0, 1\}$, and suppose some b satisfies $\tilde{x}_i^\top b > 0$ whenever $y_i = 1$ and $\tilde{x}_i^\top b < 0$ whenever $y_i = 0$.
(a) Show that $\ell(\beta) < 0$ for every β and $\ell(tb) \to 0$ as $t \to \infty$; hence no maximizer exists.
(b) Show that for $\lambda > 0$ the ridge-penalized objective $\ell(\beta) - \lambda\|\beta\|^2$ has a unique maximizer.

Solution [Supplement].

(a) With $p_i = e^{\tilde{x}_i^\top \beta}/(1 + e^{\tilde{x}_i^\top \beta}) \in (0, 1)$, each term of ℓ equals $\log p_i$ if $y_i = 1$ and $\log(1 - p_i)$ if $y_i = 0$, the log of a number in $(0, 1)$; so $\ell(\beta) < 0$. Along $\beta = tb$: if $y_i = 1$, $\tilde{x}_i^\top b > 0$ and $p_i \to 1$; if $y_i = 0$, $\tilde{x}_i^\top b < 0$ and $p_i \to 0$. Every term tends to 0, so $\sup_\beta \ell = 0$, which is never attained.

(b) $F(\beta) = \ell(\beta) - \lambda\|\beta\|^2$ is continuous, $F \le -\lambda\|\beta\|^2 \to -\infty$ as $\|\beta\| \to \infty$, so the set $\{F \ge F(0)\}$ is closed and bounded and F attains its maximum there. ℓ is concave (its Hessian is $-\tilde{X}^\top W \tilde{X}$) and $-\lambda\|\beta\|^2$ is strictly concave, so F is strictly concave and the maximizer is unique. ■

Exercise 9.2 (Logistic regression on one binary predictor)

Let $x_i \in \{0,1\}$; group x has n_x observations of which s_x have $y_i = 1$, with $0 < s_x < n_x$, and $\hat{p}_x = s_x/n_x$. Fit $\log \frac{p}{1-p} = \beta_0 + \beta_1 x$.

(a) Show that $\hat{\beta}_0 = \log \frac{\hat{p}_0}{1-\hat{p}_0}$ and $\hat{\beta}_1 = \log \frac{\hat{p}_1/(1-\hat{p}_1)}{\hat{p}_0/(1-\hat{p}_0)}$ (the log odds ratio), so the fitted probabilities are the group proportions.

(b) Show that whenever the model has an intercept, $\sum_i \hat{p}_i = \sum_i y_i$.

Solution [Supplement].

(a)

Step 1. The score equations $\tilde{X}^\top (y - p) = 0$ read $\sum_i (y_i - p_i) = 0$ and $\sum_i x_i (y_i - p_i) = 0$.

Step 2. In group x all p_i equal $p(x) = \sigma(\beta_0 + \beta_1 x)$, $\sigma(\eta) = e^\eta/(1+e^\eta)$. The second equation gives $s_1 - n_1 p(1) = 0$; subtracting it from the first gives $s_0 - n_0 p(0) = 0$. So $p(0) = \hat{p}_0$, $p(1) = \hat{p}_1$.

Step 3. Inverting σ yields $\hat{\beta}_0 = \text{logit}(\hat{p}_0)$ and $\hat{\beta}_0 + \hat{\beta}_1 = \text{logit}(\hat{p}_1)$. Both groups are present, so $\tilde{X}$ has full column rank and $W \succ 0$; the Hessian $-\tilde{X}^\top W \tilde{X}$ is negative definite, ℓ is strictly concave, and this solution is the unique MLE.

(b) The intercept's score equation is $\sum_i (y_i - \hat{p}_i) = 0$.

For the Default data with student status, $\text{logit}^{-1}(-3.5041) = 0.0292$ and $\text{logit}^{-1}(-3.0992) = 0.0431$ are the default rates of non-students and students, and $e^{0.4049} \approx 1.50$ is their odds ratio. ■

Exercise 9.3 (Reading a logistic fit; ISLR Ex. 4.6)

For students, X_1 = hours studied, X_2 = undergraduate GPA and Y = receiving an A, a logistic fit gives $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.

(a) Estimate the probability of an A for a student who studies 40 hours and has GPA 3.5.

(b) How many hours would this student need for a 50% chance?

Solution.

(a) The linear predictor is $-6 + 0.05(40) + 3.5 = -0.5$, so $\hat{p} = e^{-0.5}/(1+e^{-0.5}) = 1/(1+e^{0.5}) \approx 0.378$.

(b) $\hat{p} = 0.5$ iff the log odds are 0: $-6 + 0.05h + 3.5 = 0$, i.e. $h = 50$ hours. ■

Exercise 9.4 (Logistic versus softmax coding; ISLR Ex. 4.12)

To classify $X \in \mathbb{R}$ into apples and oranges, you fit $\Pr(\text{orange} \mid x) = e^{\beta_0 + \beta_1 x}/(1 + e^{\beta_0 + \beta_1 x})$; a friend fits the softmax model $\Pr(\text{orange} \mid x) = e^{\alpha_{o0} + \alpha_{o1} x}/(e^{\alpha_{o0} + \alpha_{o1} x} + e^{\alpha_{a0} + \alpha_{a1} x})$, both by maximum likelihood on the same data.

(a) Give the log odds of orange versus apple in both models.

(b) If $\hat{\beta}_0 = 2$, $\hat{\beta}_1 = -1$, what are the friend's estimates?

(c) If the friend gets $\hat{\alpha}_{o0} = 1.2$, $\hat{\alpha}_{o1} = -2$, $\hat{\alpha}_{a0} = 3$, $\hat{\alpha}_{a1} = 0.6$, what are yours?

(d) How often will the two predicted labels agree on 2000 test points?

Solution.

(a) Yours: $\beta_0 + \beta_1 x$. The friend's: $\log \frac{e^{\alpha_{o0} + \alpha_{o1} x}}{e^{\alpha_{a0} + \alpha_{a1} x}} = (\alpha_{o0} - \alpha_{a0}) + (\alpha_{o1} - \alpha_{a1}) x$.

Key fact. By (a), the softmax probabilities equal the logistic ones with $\beta = \alpha_o - \alpha_a$, so the friend's likelihood is $L_{\text{soft}}(\alpha) = L_{\log}(\alpha_o - \alpha_a)$. As α ranges over $\mathbb{R}^4$, $\alpha_o - \alpha_a$ ranges over all of $\mathbb{R}^2$; hence $\hat{\alpha}$ maximizes L_{soft} iff $\hat{\alpha}_o - \hat{\alpha}_a$ maximizes $L_{\log}$.

(b) Only differences are determined: $\hat{\alpha}_{o0} - \hat{\alpha}_{a0} = 2$ and $\hat{\alpha}_{o1} - \hat{\alpha}_{a1} = -1$, e.g. $\hat{\alpha}_a = (0,0)$, $\hat{\alpha}_o = (2,-1)$, or any common shift of both.

(c) $\hat{\beta}_0 = 1.2 - 3 = -1.8$ and $\hat{\beta}_1 = -2 - 0.6 = -2.6$.

(d) Always (100%): the two fits give identical probabilities at every x, hence identical labels. ■